

CS 679 Project Proposal

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Paper: [Rethinking Style Transfer: From Pixels to Parameterized Brushstrokes](#)

This paper demonstrates an extension of the paper, Image Style Transfer Using Convolutional Neural Networks, by Gatys et al. [1]. It switches the optimization space for style transfer from pixels to parameterized brushstrokes. We will reproduce the core idea using the [authors' GitHub repository](#).

The core pipeline is per-image optimization. For each (content, style) pair, we optimize a set of brushstroke parameters and render them into an image using the paper's differentiable renderer (Algorithm 1). Losses are computed using a pretrained VGG feature extractor similar to a standard neural style transfer mechanism.

The pipeline does not require supervised training data in the traditional sense. Instead, we will run optimization on a curated set of content images (e.g., 10 to 20) and style images (e.g. 3 to 5). We plan on using publicly available style images from WikiArt to reproduce stylization. We will run controlled sweeps over key hyperparameters like stroke counts and optimization steps to verify the trends described in the paper.

We will evaluate the quality of our stylized images by first qualitatively comparing them to the results obtained in the paper and by using a [pretrained model](#) to calculate the deception rate proposed by Sanakoyeu et al. [2] where feasible. We will evaluate our renderer by measuring the reconstruction MSE as a function of stroke count, and by comparing the observed trends to those reported in the paper rather than attempting to fully reproduce the large-scale benchmark proposed by Huang et al. [3].

After reproducing the baseline, we will evaluate robustness to adversarial perturbations by applying small gradient-based attacks (e.g., FGSM) to the content image before stylization and measuring degradation in reconstruction/stylization losses and visual consistency. If time permits, we will also test lightweight defenses (e.g., data augmentation, input denoising / JPEG / clipping) to see if outputs stabilize.

For computational resources, the original paper used an Nvidia TitanXP or Nvidia 2080Ti graphics card, taking around 3 minutes for brushstroke parameter optimization for each image. The method relies on iterative optimization with thousands of strokes and hundreds to thousands of gradient steps per stylization. To keep the project feasible, we will limit the scope by reducing the image resolutions or stroke counts where appropriate, decrease the number of optimization steps while preserving qualitative behavior, and minimize the

number of content-style pairs evaluated. Given our resources and time constraints, if local runtime becomes too prohibitive, we may use Google Colab GPUs for training.

References:

- [1]: Leon A. Gatys, Alexander S. Ecker, Matthias Bethge; Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 2414-2423
- [2]: Artsiom Sanakoyeu, Dmytro Kotovenko, Sabine Lang, and Bjorn Ommer. A style-aware content loss for real-time hd " style transfer. In Eur. Conf. Comput. Vis., 2018
- [3]: Zhewei Huang, Wen Heng, Shuchang Zhou; Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019, pp. 8709-8718